SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA0718 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I T.Subahan (192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks
Understanding of Concepts	25%			
Explanation	25%			
Application	20%			
Organization	15%			
Accuracy	10%			
Clarity	5%			

Total Marks /30 ***Signature of the Faculty***

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports

transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly

		control) with strong reasoning			
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

4.

Reliable data transmission is one of the primary objectives of computer networks. During communication, data may become corrupted due to electrical noise, signal interference, hardware failures, or network congestion. To ensure that the receiver gets the correct information, different layers of the OSI model work together. In this concept map, **Error Control** is achieved through the combined functions of the **Data Link Layer** and the **Transport Layer**.

The **Data Link Layer** is responsible for detecting errors that occur during transmission over a single physical link, while the **Transport Layer** ensures that the complete message is delivered correctly from the source application to the destination application. Their interaction significantly improves communication reliability.

1. Data Link Layer – Error Detection

The **Data Link Layer (Layer 2)** provides reliable communication between two directly connected devices. Its primary responsibility is **error detection**.

Functions

- Detects corrupted frames during transmission.
- Uses techniques such as:
    - **Cyclic Redundancy Check (CRC)**
    - **Checksum**
    - **Parity Bit**
- If an error is detected, the damaged frame is discarded, or a retransmission request is generated.

Example

Suppose a frame is sent from Computer A to Router B. Due to electrical interference, one-bit changes during transmission.
Original Data: 10110110
Received Data: 10111110
The CRC check detects the error and marks the frame as corrupted.
Thus, the Data Link Layer prevents incorrect data from moving to higher layers.

2. Transport Layer – Error Correction

The **Transport Layer (Layer 4)** provides **end-to-end reliability** between the sender and receiver.
Its responsibilities include:

- Error correction
- Retransmission of lost packets
- Sequence numbering
- Flow control
- Acknowledgements (ACK)
- Timeout mechanism

The most common protocol is **TCP (Transmission Control Protocol)**.

Working

- Every packet is assigned a sequence number.
- The receiver sends an acknowledgement (ACK) after receiving the packet.
- If the sender does not receive the ACK within a specified timeout period, it retransmits the packet.

This ensures that missing or corrupted packets eventually reach the receiver correctly.

3. Interaction Between the Layers

Although both layers deal with errors, they work at different levels.

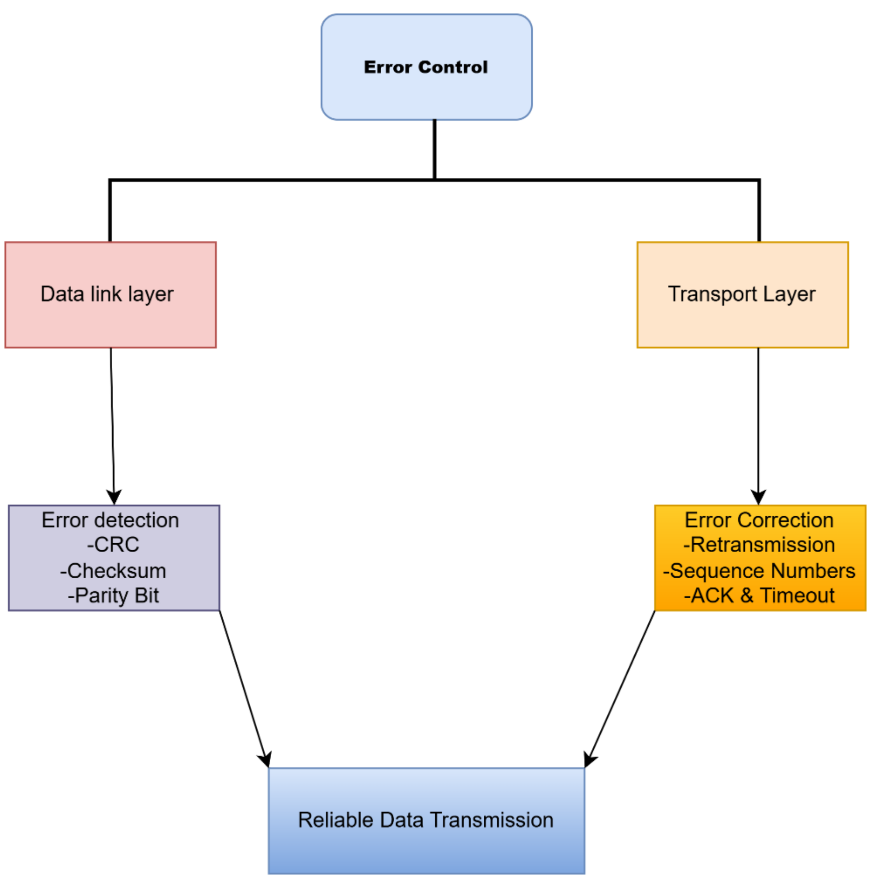
Step 1

The **Data Link Layer** first checks every frame using CRC or checksum.
↓
Step 2

If the frame is error-free, it forwards it to the Transport Layer.
↓
Step 3
If a frame is lost or corrupted and cannot be recovered locally, the **Transport Layer** detects the missing packet using sequence numbers.
↓
Step 4
The Transport Layer requests retransmission and ensures that all packets arrive in the correct order.
Together, these layers provide complete reliability.

How Their Interaction Improves Reliability
The interaction between these layers improves reliability in several ways.
Early Error Detection:
The Data Link Layer identifies transmission errors immediately before forwarding the frame.
Benefit:
Corrupted data does not propagate through the network.

Concept Map Diagram:



5.
The **OSI (Open Systems Interconnection) model** is a seven-layer networking model that explains how data travels from a user's web browser to a web server and back. During web browsing, each layer performs a specific task to ensure that the

requested webpage is delivered accurately and efficiently. If any layer fails, communication is affected, and the webpage may not load correctly.

OSI Layers in Web Browsing

1. Application Layer (Layer 7)

The Application Layer is the closest layer to the user. It provides services that allow users to access websites.

Functions

- Uses **HTTP** and **HTTPS** protocols.
- Sends webpage requests.
- Receives webpage content.
- Communicates with web browsers such as Chrome or Edge.

2. Presentation Layer (Layer 6)

The Presentation Layer formats and secures data.

Functions

- Encrypts data using **SSL/TLS**.
- Decrypts received data.
- Compresses data.
- Converts different data formats.

Example: HTTPS websites use encryption to protect passwords and banking information.

3. Session Layer (Layer 5)

This layer manages communication sessions.

Functions

- Establishes connection.
- Maintains the communication session.
- Terminates the session after communication ends.

Example: Keeps the browser connected while loading multiple webpage resources.

4. Transport Layer (Layer 4)

The Transport Layer ensures reliable communication.

Functions

- Uses **TCP**.
- Divides data into segments.
- Assigns port numbers.
- Performs error recovery.
- Uses acknowledgements and retransmission.

Example: Ensures that all webpage data arrives correctly.

5. Network Layer (Layer 3)

The Network Layer determines the best path for data.

Functions

- Uses IP addresses.
- Routes packets between networks.
- Finds the shortest communication path.

Example: Directs packets from your computer to Google's server.

6. Data Link Layer (Layer 2)

This layer provides communication between devices on the same network.

Functions

- Uses MAC addresses.
- Detects transmission errors.

- Creates data frames.

Example: Transfers data from your laptop to your Wi-Fi router.

7. Physical Layer (Layer 1)

The Physical Layer transmits raw bits.

Functions

- Uses cables, optical fiber, or Wi-Fi.
- Converts data into electrical, optical, or radio signals.

Example: Wi-Fi signals carrying webpage data.

How Failure in One Layer Affects the Entire Communication Process

Each OSI layer depends on the services of the layer below it. Therefore, a failure in any one layer can interrupt the entire web browsing process.

1. Application Layer Failure

- Browser cannot send HTTP/HTTPS requests.
- Website cannot be accessed.

Result: Webpage does not open.

2. Presentation Layer Failure

- SSL/TLS encryption fails.
- Secure connection cannot be established.

Result: Browser displays security warnings such as **"Your connection is not private."**

4. Transport Layer Failure

- TCP cannot guarantee reliable delivery.
- Packets may be lost or arrive out of order.

Result: Slow loading, incomplete webpage, or failed downloads.

5. Network Layer Failure

- Incorrect IP addressing or routing.
- Packets cannot reach the destination server.

Result: Browser shows **"Server not found"** or **"Network unreachable."**

6. Data Link Layer Failure

- Frame errors occur.
- MAC communication fails.

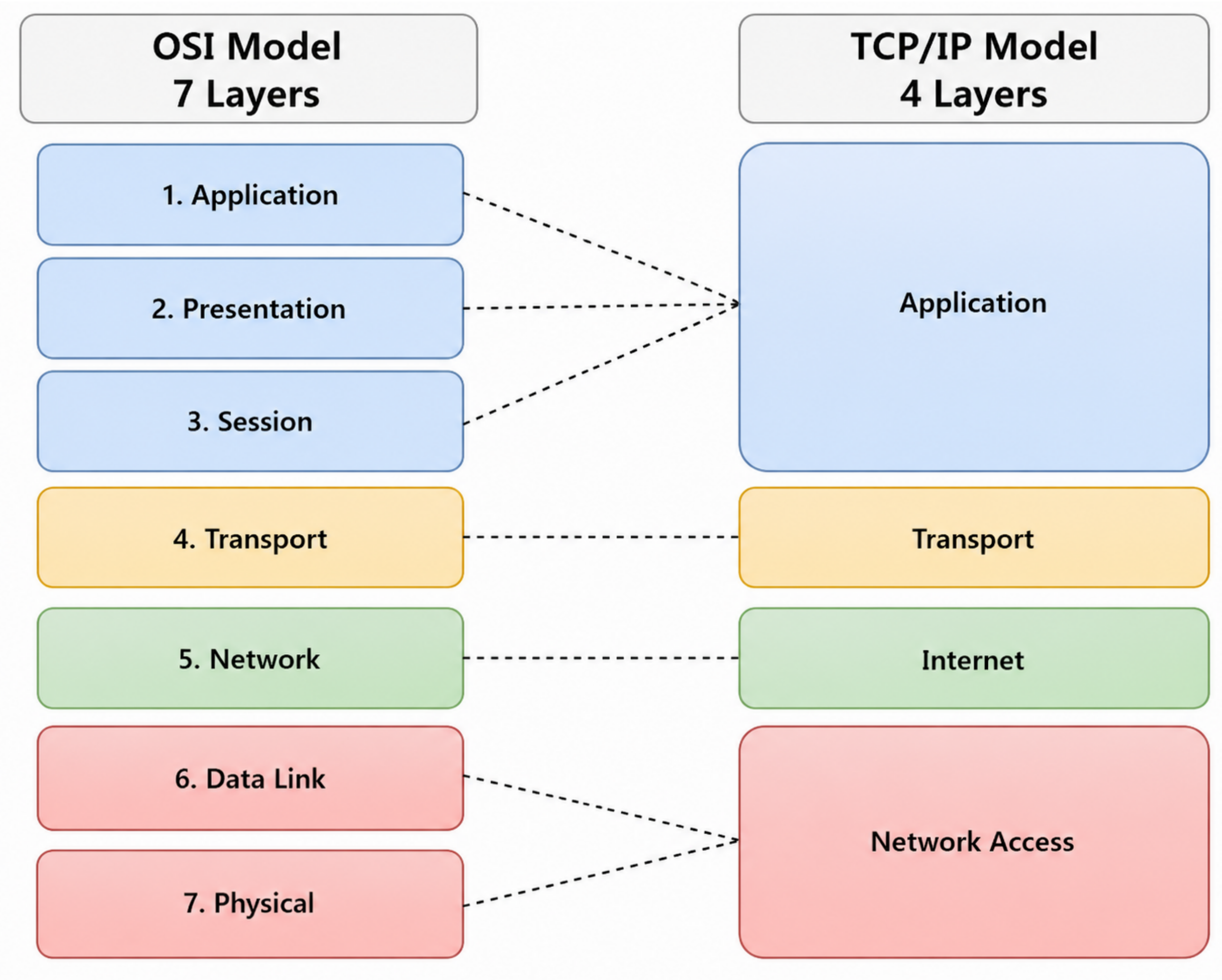
Result: Frequent packet loss, unstable connection, or inability to communicate with the router.

7. Physical Layer Failure

- Damaged cable or Wi-Fi signal interruption.
- No signal transmission.

Result: Complete loss of internet connectivity.

Concept Map Diagram:


OSI Model
7 Layers
TCP/IP Model
4 Layers
1. Application
2. Presentation
3. Session
Application
4. Transport
Transport
5. Network
Internet
6. Data Link
7. Physical
Network Access

6.

The **OSI (Open Systems Interconnection) model** is a seven-layer networking framework used to understand how data travels through a network. It is also an effective troubleshooting tool because it helps identify the exact layer where a problem occurs.

Analysis of the Problem

The concept map shows that the **Physical Layer** and **Data Link Layer** are working correctly.

This means:

- Network cables or Wi-Fi are functioning.
- Devices are powered on.
- MAC address communication is successful.
- Frames are transmitted correctly between directly connected devices.

However, the **Network Layer** has failed.

The Network Layer is responsible for:

- IP addressing
- Packet routing
- Finding the destination path
- Delivering packets across different networks

Since the Network Layer is not working, packets cannot reach the destination device. As a result, the **Transport Layer** cannot establish a connection, and the **Application Layer** cannot access services such as websites or email.

Layer-by-Layer Analysis

1. Physical Layer

Status

Working correctly.

Functions

- Transfers electrical, optical, or wireless signals.
- Uses Ethernet cables, fiber optics, or Wi-Fi.

Meaning

Since this layer is working:

- Cable connections are proper.
- Wi-Fi signal is available.
- Hardware devices are powered on.

2. Data Link Layer

Status

Working correctly.

Functions

- Uses MAC addresses.
- Detects frame errors.
- Transfers data between neighboring devices.

Meaning

The computer can successfully communicate with the local switch or router.

3. Network Layer

Status

Failed.

Functions

- IP addressing
- Routing packets

- Selecting the best path

Possible Causes

- Incorrect IP address
- Wrong subnet mask
- Incorrect default gateway
- Router failure
- Routing table error
- IP conflict

Effect

Packets cannot travel beyond the local network.

4. Transport Layer

Status

Failed because the Network Layer has failed.

Functions

- TCP communication
- Reliable delivery
- Port numbers
- Error recovery

Effect

Without IP connectivity:

- TCP connection cannot be established.
- Packets cannot be acknowledged.
- Data transfer stops.

5. Application Layer

Status

Failed.

Functions

- Web browsing
- Email
- File transfer
- Video streaming

Effect

Applications cannot communicate because the lower layers cannot deliver the data.

How the Failure Affects the Entire Communication Process

The communication process stops at the **Network Layer**.

The sequence is:

Physical Layer
✔
↓
Data Link Layer
✔
↓
Network Layer
✖
↓
No IP Routing
↓
Transport Layer Cannot Establish TCP
↓
Application Cannot Access Services

Since packets cannot be routed, higher-layer protocols have no path to communicate.

How the Concept Map Helps in Troubleshooting
A concept map visually represents the relationship between OSI layers and makes troubleshooting easier.

1. Identifies the Fault Location
The concept map clearly shows that the first failed layer is the **Network Layer**.
This allows the network administrator to focus on:

- IP configuration
- Router settings
- Routing tables
- Gateway configuration

instead of checking cables or switches unnecessarily.

2. Saves Time
Instead of testing every networking component, the administrator immediately knows:

- Physical hardware is working.
- Data Link communication is working.
- The problem starts at Layer 3.

This speeds up troubleshooting.

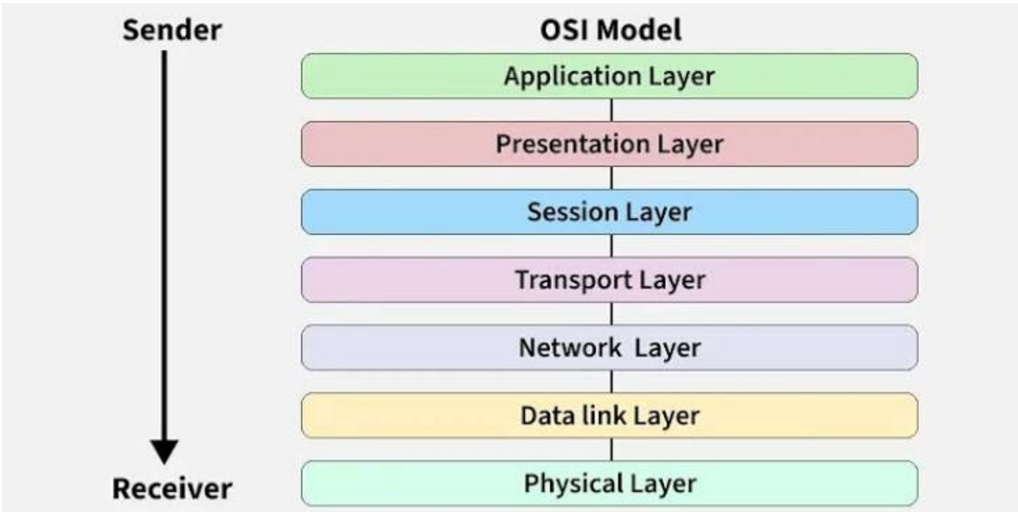
3. Shows Layer Dependency
The concept map demonstrates that higher layers depend on lower layers.
Since the Network Layer fails:

- Transport Layer cannot establish TCP connections.
- Application Layer cannot access network services.

This helps explain why multiple services stop working even though the hardware is functioning.

4. Improves Problem Analysis
The concept map makes it easier to understand the flow of communication and identify where the interruption occurs. It helps students and network engineers visualize how a single layer failure can affect the entire communication process.

Concept Map Diagram:


Sender
OSI Model
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data link Layer
Physical Layer
Receiver